

Fractal Trees

While the artificial fractals such as the Sierpiński triangle and carpet are perfectly self-similar, fractals can include shapes that do not have perfect self-similarity. Fractal geometry, as envisioned by mathematician Benoit B. Mandelbrot (whose middle initial recursively stands for Benoit B. Mandelbrot) included natural shapes such as mountains, coastlines, plants, blood vessels, and the clustering of galaxies as fractals. Upon close examination, these shapes continued to consist of “rougher” shapes not easily contained by the smooth curves and straight lines of simplified geometry.

As an example, we can use recursion to reproduce *fractal trees*, whether perfectly or imperfectly self-similar. Generating trees requires creating a branch with two child branches that issue from their parent at set angles and decrease at set lengths. The Y shape that they produce is recursively repeated to create a convincing drawing of a tree, as in Figures 9-11 and 9-12.